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Experiment 5B (Blue Dye 2)

1. The report proceeds from the following assumptions:

   (a) "Blue Dye No. 1" refers to the Brilliant Blue FCF industrial colorant (this has the closest maximum absorbance wavelength to the tested blue dye — both have maximum absorbances at 628 nm).

   (b) The concentration of Brilliant Blue FCF in "Cool Blue" is equivalent to the concentration of the mystery blue dye found experimentally (0.02 g/L = 20 mg/L).

   (c) A single unit of "Cool Blue" has nominal volume 350 mL.

   (d) The maximum ADI (acceptable daily intake) for the colorant in "Cool Blue" is 6 mg per kilogram of body weight per day (based on the World Health Organization's 2017 max ADI-value for Brilliant Blue FCF)

REPORT

1. We find the general form of the dosage administered in N units of "Cool Blue" over the time interval t.

Let w be the weight of the consumer, in kilograms
Let t be the consumption interval, in days
Let N be the number of "Cool Blue" units consumed
Let C be the concentration of the "Cool Blue" colorant, in milligrams per Liter
Let V be the nominal volume of a single unit of "Cool Blue", in Liters
Let d be the per-kilogram-per-day dosage of blue dye

Then the dosage-per-kilogram-day is

$$d(w,t,N,C,V) = \frac{NCV}{wt} \qquad \left(\frac{\text{mg}}{\text{kg}\cdot\text{day}}\right)$$

For our particular scenario, take $C = 0.0200$ g/L = **20** mg/L and $V = 0.35$. Then the dosage-per-kilogram-day is

$$d(w,t,N) = \frac{7N}{wt} \qquad \left(\frac{\text{mg}}{\text{kg}\cdot\text{day}}\right)$$

The ADI is assumed to be 6 mg per kilogram of body weight per day (World Health Organization). Hence, **the drink is safe for all** (w, t, N) **in the subset** $U \subset \mathbb{R}^3$ **such that** $d(w, t, N) \leq 6$.

If we standardize this dosage to $w = 90$ kg and $t = 1$ day, then

$$d(90, 1, N) = \frac{7N}{90} \leq 6 \implies N \leq 77.143$$

Hence, we can say that the average American male should consume no more than **77** units (7.12 gallons) of "Cool Blue" per day to stay under the ADI limit. Since this limit exceeds what a regular person could reasonably consume on a daily basis, this concentration of "Cool Blue" is declared **safe to consume**.

2. Let $A(C)$ be the absorbance at concentration C.

$$\begin{aligned}
\text{Blowing last dye out} &\implies V_{\text{dye, recorded}} < V_{\text{dye, actual}} \\
&\implies C_{\text{recorded}} < C_{\text{actual}} \\
&\implies A(C_{\text{recorded}}) < A(C_{\text{actual}}) \\
&\implies \text{but we plot } (C_{\text{recorded}}, A(C_{\text{actual}})) \text{ on the calibration curve} \\
&\implies \text{every non-zero point moves vertically } A(C_{\text{actual}})) - A(C_{\text{recorded}}) > 0 \\
&\implies \text{The blank still sits at } (0, 0) \text{ presumably} \\
&\implies \text{The calibration curve slope } m \text{ increases} \\
&\implies \text{The calculated concentration } \frac{A(C_{\text{unknown}}) - b}{m} \text{ decreases}
\end{aligned}$$

Note that you can't mess up the blank by blowing the last dye out because there's no dye to blow out in the first place. Hence, the calibration curve still must go through $(0, 0)$. The concentration of the unknown would be **falsely low**.

Because the concentration of the unknown would be reported falsely low, the intake risk assessment would become falsely milder; the projected allowable daily intake would become larger than what it should have been, causing a safety issue. If the concentration of the unknown were reported falsely high, however, there would have been no safety issue (the risk assessment would have been more stringent than required).

Bibliography

1 Joint FAO/WHO Expert Committee on Food Additives, "Brilliant Blue FCF," WHO JECFA Database, 2017. Accessed: Oct. 24, 2026. [Online]. Available: https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/3309